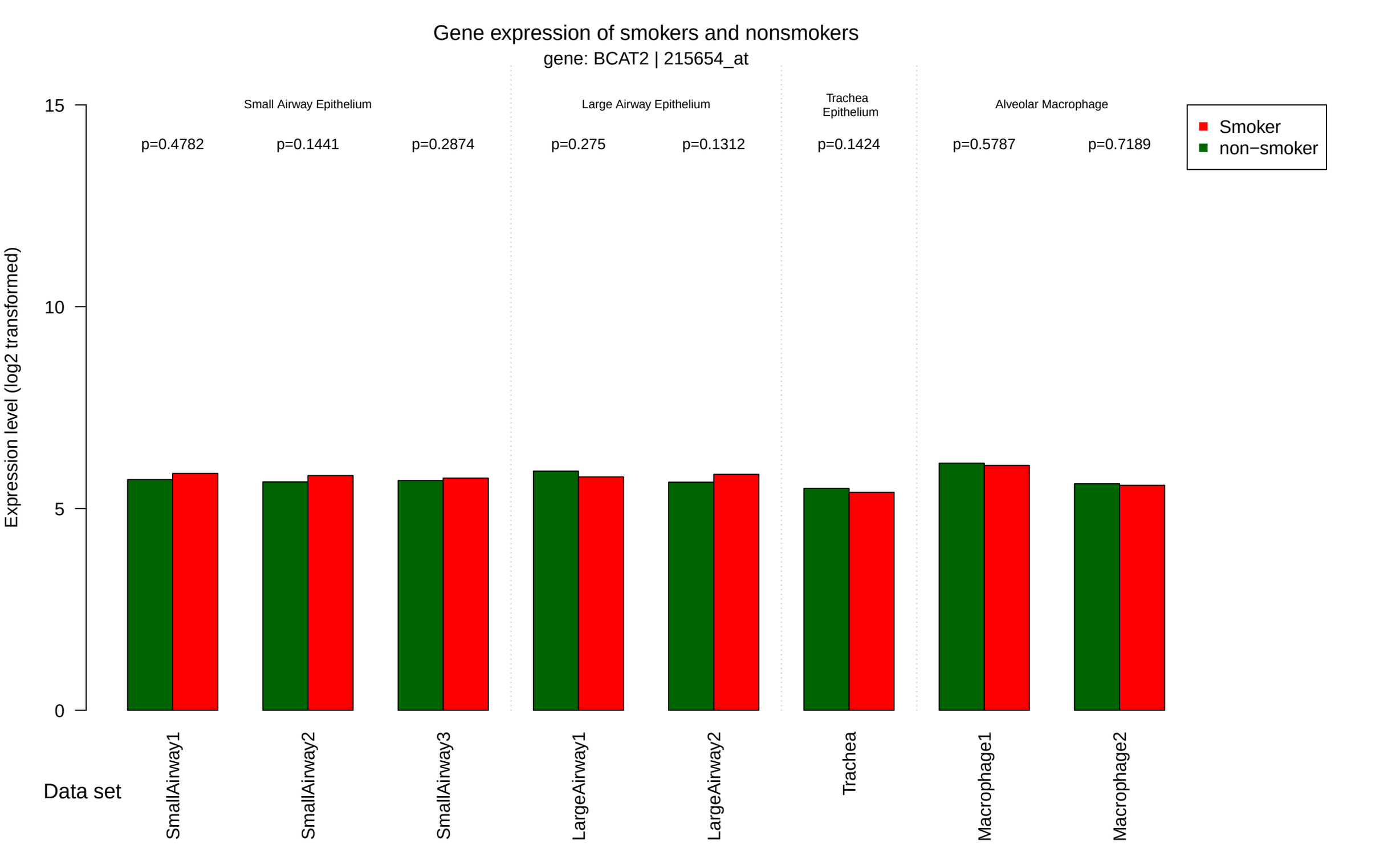
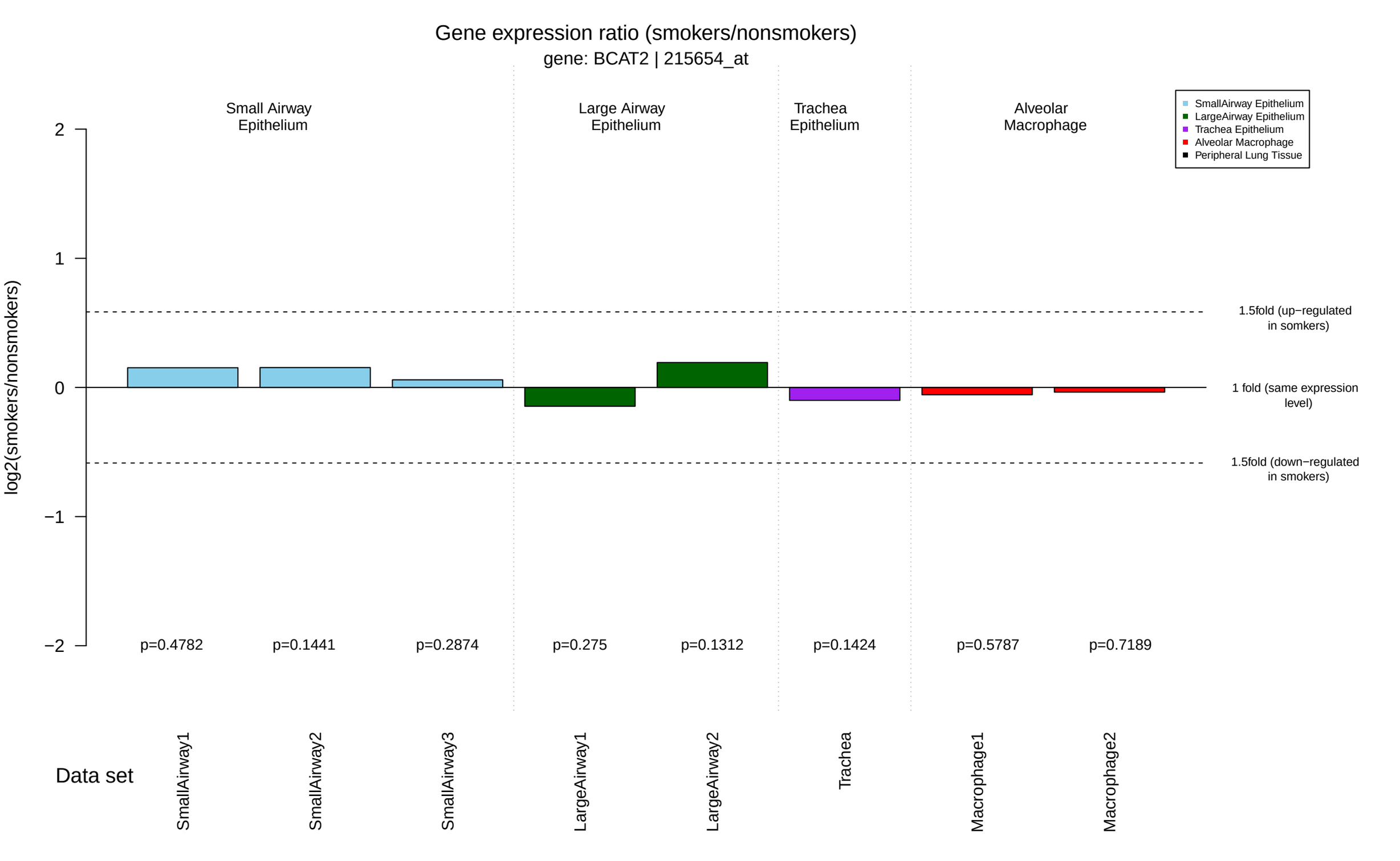
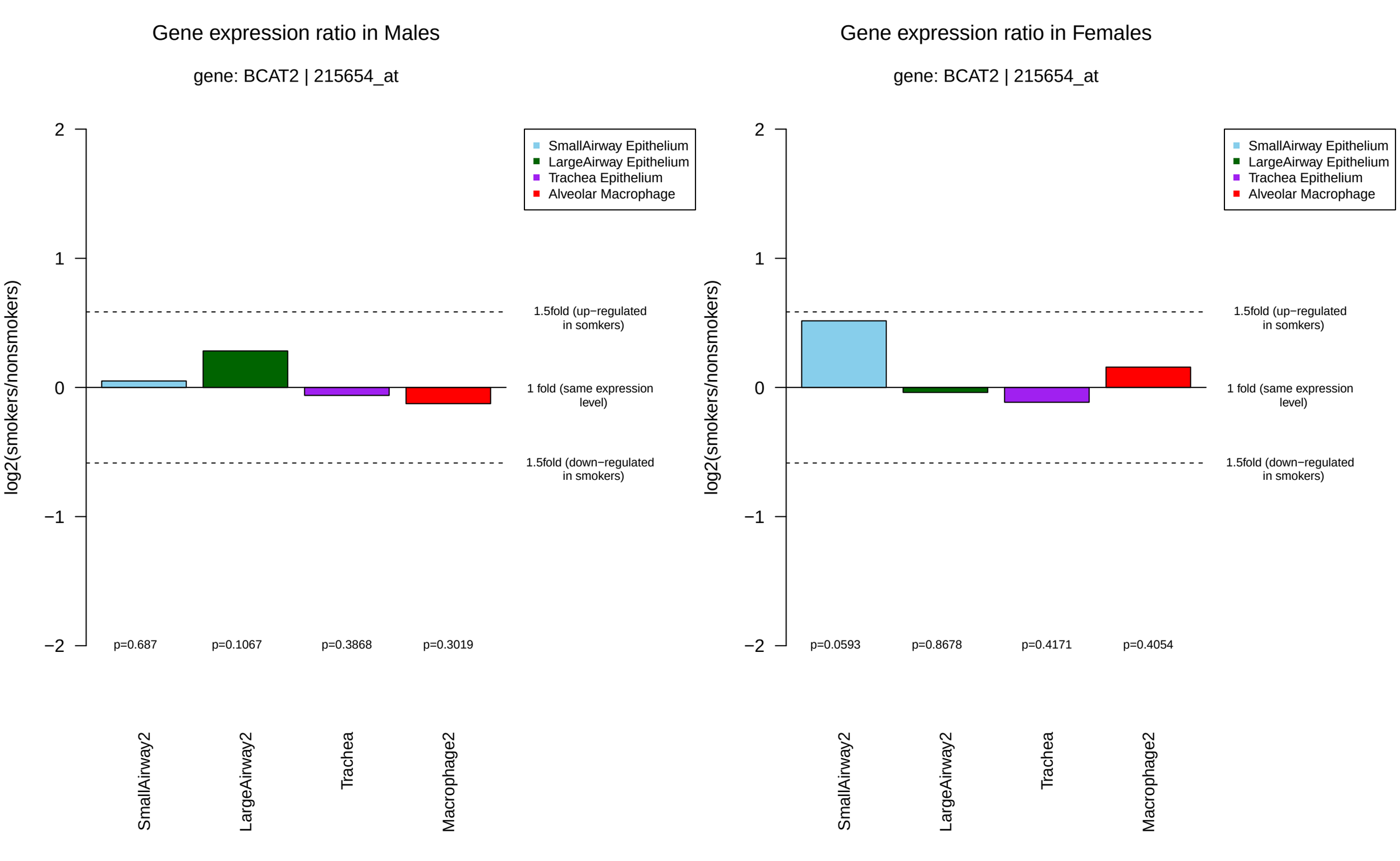
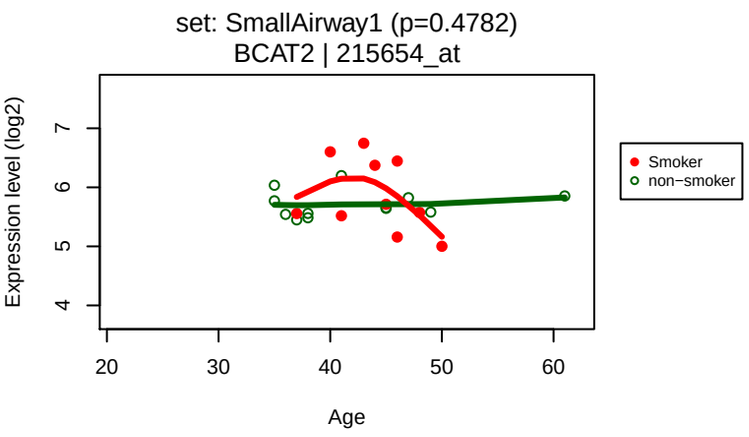
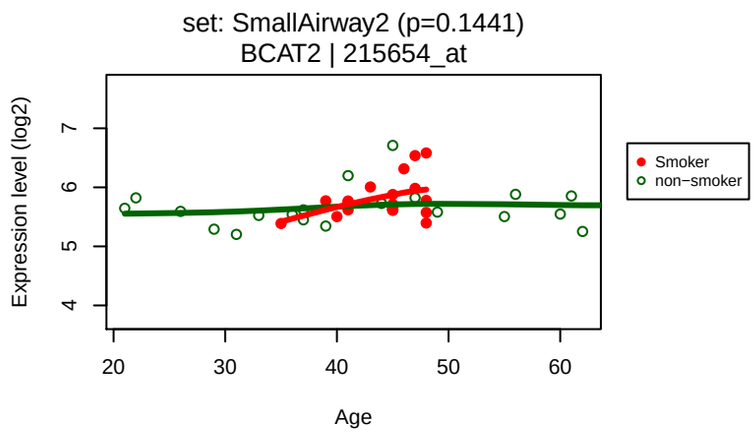
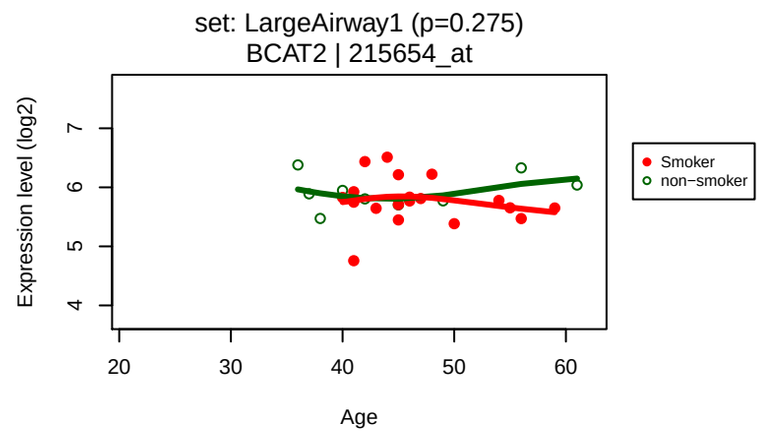

Gene expression of smokers and nonsmokers
gene: BCAT2 | 215654_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.4782
p=0.1441
p=0.2874
p=0.275
p=0.1312
p=0.1424
p=0.5787
p=0.7189
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

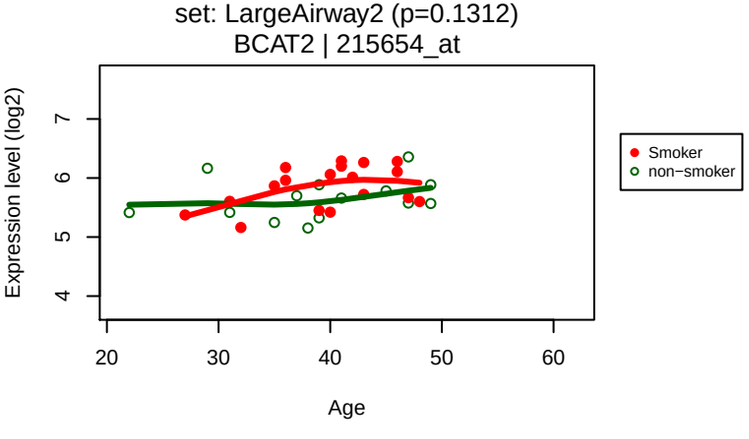

Gene expression ratio (smokers/nonsmokers)
gene: BCAT2 | 215654_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4782
p=0.1441
p=0.2874
p=0.275
p=0.1312
p=0.1424
p=0.5787
p=0.7189
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

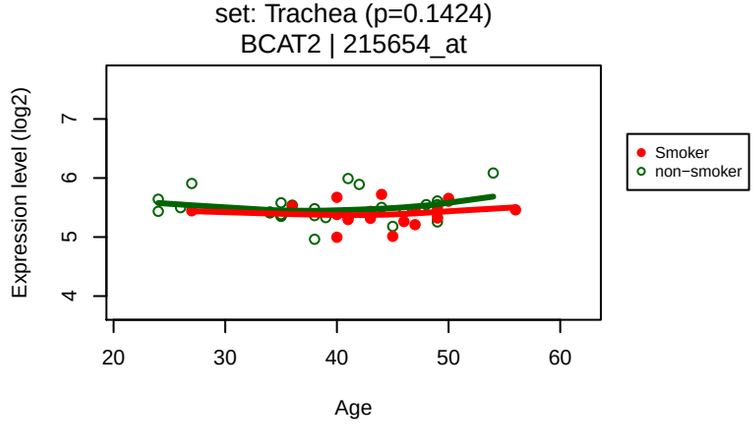

Gene expression ratio in Males
gene: BCAT2 | 215654_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.687
p=0.1067
p=0.3868
p=0.3019
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: BCAT2 | 215654_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0593
p=0.8678
p=0.4171
p=0.4054
SmallAirway2
LargeAirway2
Trachea
Macrophage2

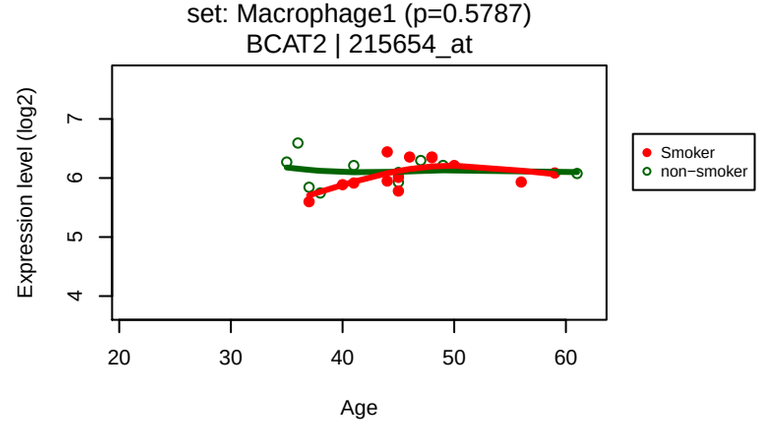

set: SmallAirway1 (p=0.4782)
BCAT2 | 215654_at
Expression level (log2)
Age
Smoker
non−smoker

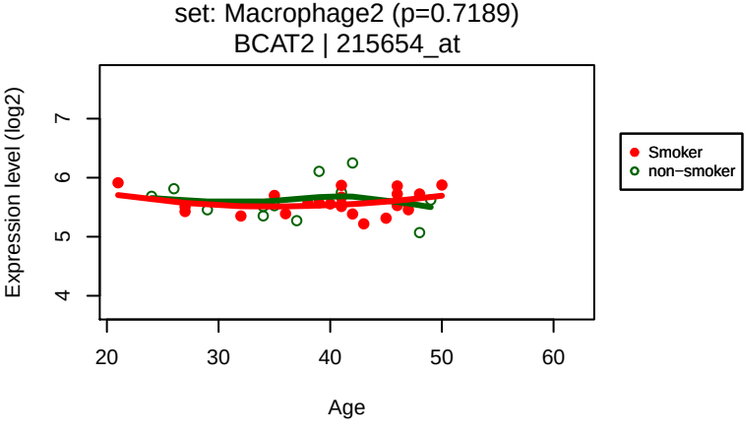

set: SmallAirway2 (p=0.1441)
BCAT2 | 215654_at
Expression level (log2)
Age
Smoker
non−smoker

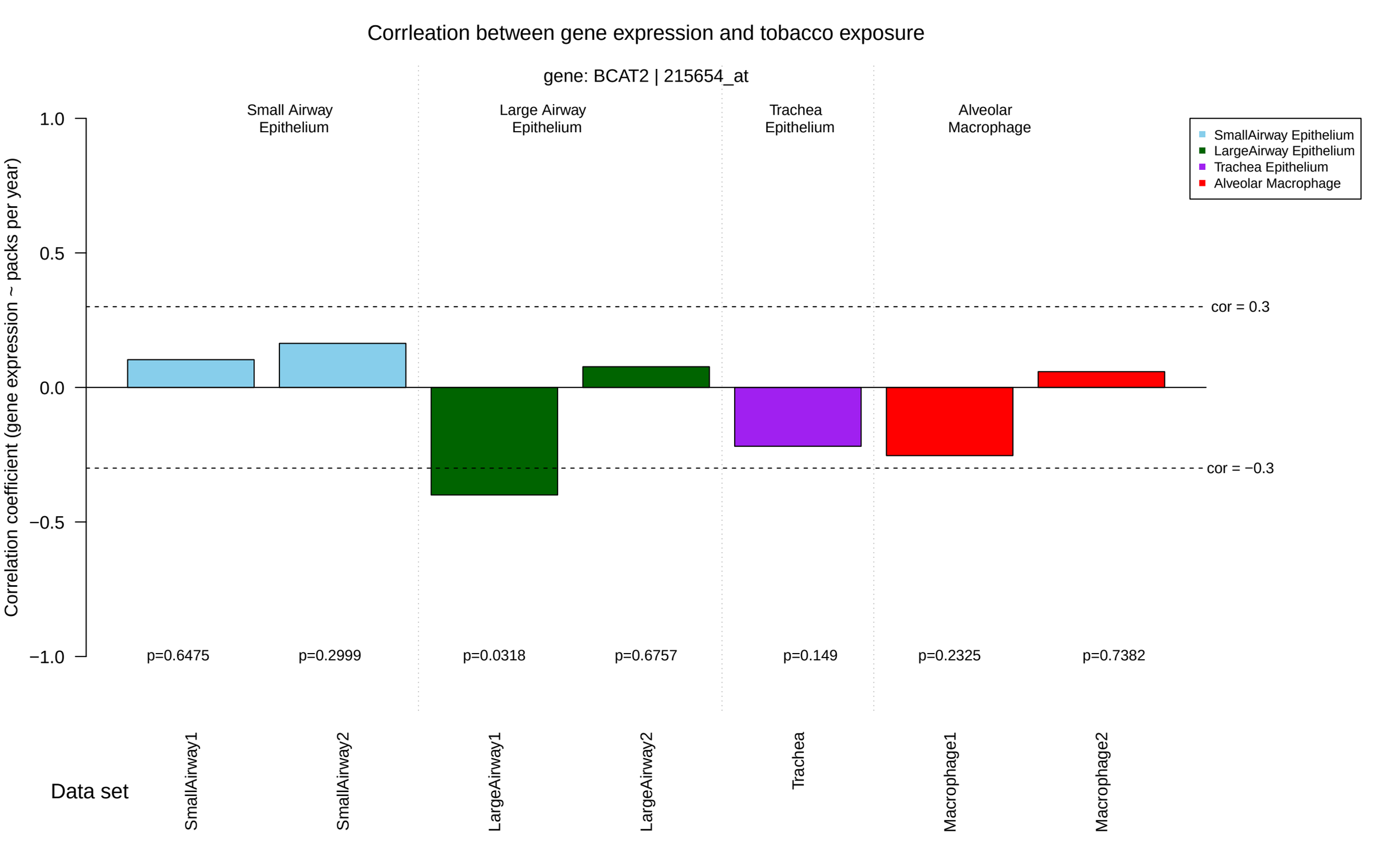

set: LargeAirway1 (p=0.275)
BCAT2 | 215654_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.1312)
BCAT2 | 215654_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.1424)
BCAT2 | 215654_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.5787)
BCAT2 | 215654_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.7189)
BCAT2 | 215654_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: BCAT2 | 215654_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.6475
p=0.2999
p=0.0318
p=0.6757
p=0.149
p=0.2325
p=0.7382
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2